

Dikshant Sagar

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AI Researcher and Machine Learning Engineer with 8+ years of experience developing and scaling deep learning and foundation models for scientific discovery, currently pursuing a Ph.D. in Computer Science at UC Irvine. Published researcher with 15+ peer-reviewed papers spanning biological modeling, physics-guided generative models, and large-scale scientific AI systems. Experienced in training and fine-tuning large vision-language models (VLMs/LLMs) and diffusion models, designing agentic reasoning systems, and building end-to-end data and evaluation pipelines for biological and physical sciences applications. Strong background in large-scale GPU training (DeepSpeed, Accelerate, CUDA), multimodal learning, and cross-disciplinary collaboration with domain scientists. Passionate about building frontier AI systems that enable biological reasoning, hypothesis generation, and open scientific discovery.

Education

University of California, Irvine Ph.D. in Computer Science [GPA: 4.0] Advisor: <i>Dr. Pierre Baldi</i> Research Areas: <i>Machine Learning, Deep Learning, AI for Healthcare, Biomedicine, and Physics.</i>	2024-Present
California State University, Los Angeles M.S. in Computer Science [GPA: 4.0]	2022-2024
Indraprastha Institute of Information Technology, Delhi B.Tech in Computer Science	2017-2021

Skills

Programming: Python, Javascript, C/C++, Java, SQL
ML/DL: PyTorch, Transformers, WandB, PyTorch Lightning, Scikit-Learn, TensorFlow, Keras, OpenCV
Methods: Vision-language models, LLM fine-tuning (QLoRA), diffusion models, multimodal learning, self-supervised learning, RAG, multi-agent systems, efficient inference
Systems: DeepSpeed, Accelerate, CUDA, Slurm/HPC, Docker, Git, Linux, WandB, AWS, GCP

Research Experience

Graduate Research Assistant, Baldi Lab, UC Irvine - Developing Pixel-space diffusion denoising trajectories for hyperspectral cellular imaging to enable unsupervised organelle segmentation and longitudinal tracking within MDA-MB-231 breast cancer cells. - Implementing a Self-supervised learning methods for FLAME multiphoton skin imaging to learn robust representations for automated differentiation and localization of melanoma and other skin lesions, with supervised classification and Grad-CAM providing complementary baselines and interpretability. - Engineering a Multimodal temporal learning model integrating physiological signals (HR, HRV, EDA, sleep), wearable sensor data, and patient-reported outcomes for short-horizon prediction of pediatric postoperative pain. - Architecting a Joint vision-language foundation model for neutrino event analysis by extending LLaMA 3.2-11B Vision-Instruct with a multi-scale segmentation head; enabling unified event classification, chain-of-thought explanation, and particle-level semantic mask prediction within a single model. - Designing an end-to-end Multi-Agentic AI framework for NOvA and DUNE DAQ Operations (Fermilab) that autonomously detects, diagnoses, plans, and executes recovery for scientific detector anomalies by combining multimodal anomaly detection, multi-agent RAG, and operational knowledge from logs, documentation, code, and historical incidents. - Devised a Hyperspectral-imaging CNN to identify chemoresistance in MDA-MB-231 cells from phasor unmixed cell images, achieving 94% accuracy/AUROC and using 21 organelle-pair ablations; we identify which organelles are most deterministic of chemoresistance. - Fine-tuned LLaMA 3.2-11B Vision-Instruct LLM with QLoRA for neutrino event classification and explanation; improved baseline accuracy by 28%. - Developed Sparsity invariant frame interpolation for tilt series generation in Scanning/Transmission electron microscopy; improved PSNR by 6.6% and SSIM by 3.3% over the baseline. - Implemented Scalable Deep Learning based Signal Processing of LArTPC for efficient neutrino track segmentation inference on CPU; improved memory usage by 84% and time usage by 87% while maintaining accuracy. - Optimized and Improved RNA-Seq differential analysis pipeline using HISAT, Cufflinks and CyberT, achieving 10× performance boost. - Created a Deep learning model for multi-slice melanocyte segmentation and PRAME index prediction, reducing PRAME IHC orders by 30%, cutting TAT by 3.5 min/case, and saving around \$28K/1K cases annually at fixed sensitivity.	2024-Present
Graduate Research Assistant, Computational Molecular Biology Lab, CSULA Built Physics-guided deep generative models for conditional de novo structure based drug discovery; <i>CSBW (ACM-BCB) 2023 Best Paper Finalist</i>; Improving average binding affinities by 13.4% from previous state of the art method.	2023-2024
Graduate Research Assistant, Machine Learning and Sensing Lab, CSULA	2023-2024

- Proposed dual-encoder self-attention model for **GABA Edited-MRS reconstruction**; achieved 95% lower MSE and 450% higher SNR vs. baseline (simulated raw MRS data).

Research Associate, SBI Lab, IIIT Delhi

2021-2022

- Created multi-head self-attention-based **Representation learning for unsupervised clustering of flow cytometry data**; Improving by 14% on average on six benchmark datasets over the previous state of the art.
- Developed **Hybrid GCN (GCRS)** for multiple myeloma risk stratification; published in *Computers in Biology and Medicine* (IF 7.7).

Undergraduate Researcher, MIDAS Lab, IIIT Delhi

2020-2020

- Built **Deep Learning-based attribute-wise user preference matching** and learning using Bayesian Personalized Ranking, obtaining an AUC of 0.8502, increasing it by 2% more than the best previous work.

Selected Publications

- **Sagar, D.**, Yu, K., Yankelevich, A., Bian, J., & Baldi, P. *Adapting vision-language models for neutrino event classification in high-energy physics*. Communications Physics, 2026
- Robles E., **Sagar D.**, et al. *Heterogeneous Point Set Transformers for Segmentation of Multiple View Particle Detectors* Accepted to NeurIPS 2025 Workshop - Machine Learning for the Physical Sciences.
- **Sagar D.**, et al. *Angular Sparsity Invariant Tilt Series Generation in Scanning/Transmission Electron Microscopy* Accepted to NeurIPS 2025 Workshop - Machine Learning for the Physical Sciences.
- **Sagar D.**, et al. *Scalable Inference for LArTPC Signal Processing with MobileU-Net and Overlap-Tile Chunking* Accepted to NeurIPS 2025 Workshop - Machine Learning for the Physical Sciences.
- Gupta A., Hooda R., Motwani S., **Sagar D.**, et al. *MuSARCyto: Multi-Head Self-Attention for Cytometry Data Clustering*. Cytometry A, 2025.
- **Sagar D.**, Jasko A., Forouzesh N. *Enhancing Drug Discovery via Physics-Guided Deep Generative Models*. Comp. Struct. Bioinformatics Workshop, 2024.
- Chatterjee R., **Sagar D.**, et al. *Deep Residual Distilled CNNs for Stroke Detection*. International Journal of Semantic Computing, 2024.
- **Sagar D.**, Risheh A., Sheikh N., Forouzesh N. *Physics-Guided Deep Generative Model for Ligand Discovery*. ACM-BCB, 2023. **Best Paper Finalist**.
- **Sagar D.**, Aggarwal P., Farswan A., Gupta R., Gupta A. *GCRS: Hybrid Graph Convolutional Network for Risk Stratification in Multiple Myeloma*. Computers in Biology and Medicine, 2022.

Publications (Under Review)

- **Sagar D.**, et al. *Explanation-Guided Fine-Tuning of Vision-Language Models for Neutrino Interaction Prediction and Human-in-the-Loop Evaluation* Under review in Nature Machine Intelligence, September 2026.

Awards & Honors

- UCI AGS Travel Grant (2025, 2026).
- UCI Computer Science Department Research Fellowship (2024).
- Outstanding Graduate Student Award, CSULA (2024).
- Outstanding Oral Presentation, RSCA (2024).
- Best Paper Finalist, ACM-BCB (2023).
- CSUBIOTECH Graduate Scholarships & Travel Grants (2022-2024).

Teaching

- Teaching Assistant, COMPSCI 274 - Neural Networks and Deep Learning, UC Irvine (Winter 2026, Spring 2026)
- Teaching Assistant, COMPSCI 171 - Introduction to Artificial Intelligence, UC Irvine (Winter 2025, Spring 2025).
- Teaching Assistant, CS4550 - Computer Graphics, CSULA (Fall 2023).
- Teaching Assistant, CS4540 - Data Visualization, CSULA (Fall 2022).
- Instructor, Intro to Deep Learning, CSULA (Summer 2023, Summer 2024).

Academic Service

Reviewer

- Machine Learning and the Physical Sciences Workshop (2026)
- IEEE Transactions on Computational Biology and Bioinformatics (TCBB 2026)
- Machine Learning and the Physical Sciences Workshop (NeurIPS 2025)
- Neural Networks Journal (2024)